

Selection Beats Sophistication: Discovering Interpretable Creativity Metrics with Adversarial Agent Evolution

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Abstract

Judging whether one piece of writing is more *creative* than another is easy for people and surprisingly hard to automate with anything transparent. Large language models can act as judges, but they are slow, expensive, and opaque. We ask whether a small team of language-model *agents* can instead **discover** short, fast, fully inspectable scoring functions — ordinary Python that anyone can read — that rank a fresher rewrite of a text above a flatter one. Our system pits two agent roles against each other: an *engineer* that writes new scoring functions after studying where the current best one fails, and an *adversary* that writes new test cases designed to fool the current best. A deterministic referee filters every proposal through four automatic gates: it must behave correctly, must not memorize the test set, must behave differently from every existing scorer, and must genuinely separate better rewrites from worse ones. Running this loop, the system rediscovered — without being told — a classical principle from creativity research: a good creativity measure must combine a novelty signal with a sanity check, multiplicatively. We then ran the same loop with a much smaller model generating three times as many proposals. The smaller model's individual ideas were worse, yet the loop converged to an equally strong final scorer. Acceptance rates were identical under both models, and both runs converged on the same design shape, suggesting that the selection pressure — not the sophistication of the proposing model — is the primary driver of discovery. We report the full trajectories, the failure modes our honesty checks caught (including a champion that secretly overfit the test distribution and a test-set-growth rule that backfired), and the ceiling we hit: word-statistics can measure freshness, but judging whether a fresh image actually *fits* still requires meaning-aware tools.

1 Introduction

Suppose an editor rewrites the sentence “*The storm arrived late at night*” as “*Night cracked open into a storm.*” Most readers agree the second is more creative. Now suppose we want a program to make that call — millions of times, cheaply, and in a way we can audit. This is the problem of **scoring subjective text quality**, and creativity is among its hardest instances.

The fashionable answer is to ask a large language model (LLM) to be the judge. That works, but the judge is a black box: we cannot inspect *why* it preferred one text, we cannot run it offline in microseconds, and its preferences drift with the model version. The classical answer is to hand-craft a metric — count rare words, count clichés — but hand-crafted metrics encode their author's blind spots and, as we show, collapse the moment a test case decouples the proxy from the quality it stands for.

This paper explores a third path: let LLM agents **write the metric as code**, and let an evolutionary process — selection on measured performance, plus an adversary that continually manufactures harder test cases — decide what survives. The output is not a neural network but a short, deterministic Python function anyone can read. Two questions drive the work:

Q1. Can such a loop *discover* a genuinely novel, robust scoring principle, rather than just re-tuning known ones?

Q2. When it succeeds, what deserves the credit — the intelligence of the model writing the proposals, or the evolutionary scaffolding around it? We test this directly by re-running the identical loop with a far smaller model given three times the proposal budget.

Our answers, in brief: yes (the loop independently reinvented the field's standard two-factor definition of creativity as an executable program), and — more surprisingly — **the scaffolding carries most of the weight**. The smaller model's proposals were individually weaker and more repetitive, but the acceptance rate after filtering was identical, and its final champion matched or beat the larger model's on every honest measure we track.

2 The Task, in Plain Terms

What a scorer is. A scorer is a Python function `score(text, original)` returning a number between 0 and 1. It sees a rewrite and the original it came from, and should give higher numbers to rewrites that are more creative. It must be deterministic (same input, same output), must not call any network or model, and may use only ordinary computation plus one public resource: a table of how frequent each English word is in large text corpora.

What a test case is. Each test case is a triple: an original two-sentence passage, a rewrite that a careful human would call *more* creative, and a rewrite they would call *less* creative — all describing the same facts. A scorer handles the case correctly if it gives the more creative version the higher score.

How we measure success. Accuracy (fraction of cases ranked correctly) hides too much: a scorer that squeaks past by 0.001 on every case looks identical to one that separates confidently. We therefore score both versions of every case individually and use the **separation**: the average score of all the better rewrites minus the average score of all the worse ones. A separation of 0 means the scorer cannot tell them apart at all; 1 is a perfect, maximally confident split. We report accuracy and the two averages alongside it, so over-

and under-confidence are both visible.

Why normalization matters. Every scorer must map to the same 0–1 scale, but calibrating that scale on the test cases themselves would be cheating. Scorers instead calibrate against fixed reference statistics computed once from the public word-frequency table (for example, how surprising the average English word is), so the scale is anchored to the language, not to our data.

3 Method: an Adversarial Evolutionary Loop

The system has one deterministic referee (ordinary code, no model) and two agent roles, each played by an LLM with its own private context.

The engineer proposes new scorers. Each generation it receives: a map of creativity concepts distilled from the research literature (which concepts existing scorers already measure and which are unclaimed), a statistics table for every current scorer, a similarity matrix showing which scorers behave alike, the champion's code, and the specific training cases the champion gets wrong. It must write complete, runnable modules — reasoning first, code second.

The adversary proposes new test cases. It sees only the champion's code and a list of attack angles already used, and must invent a *new* way to decouple the champion's signal from real creativity — for example, if the champion rewards rare words, write flat rewrites full of stiff formal vocabulary (high signal, no creativity) and brilliant rewrites made of everyday words (low signal, high creativity). Candidate cases are kept only if the champion actually struggles on them; most join the training set and a fixed share is diverted, sight unseen, into a held-out set.

Four acceptance gates. Every proposed scorer passes through automatic checks before it may join the population:

1. **Validity** — it runs without error, stays within 0–1, is deterministic, and is not near-constant.
2. **No memorization** — the referee parses the code and inspects every hard-coded word list; if it contains rare words copied from the visible test cases, or any large content word list at all, the scorer is rejected. (This gate exists because, in a pilot, an agent shown failing cases quietly copied eleven of their distinctive verbs into its feature list, inflating its measured skill.)
3. **Behavioral novelty** — we record, for every scorer, its per-case score difference between the better and worse rewrite. If a new proposal's pattern of differences correlates above 0.6 with any existing scorer's, it is the same signal in new clothes and earns no novelty credit, no matter how different its stated mechanism sounds.
4. **Usefulness** — it must beat the population's median separation, or be genuinely novel with positive separation.

Three tiers of test data keep everyone honest. The *training* cases are visible to agents and drive selection. The *held-out* cases are never shown to any agent and exist only

so we can check whether a champion's skill is real. The *stationary core* is a small, never-growing subset of the original cases, giving one number that stays comparable across all generations even as the test set grows harder.

Schedule. The population holds at most ten scorers: the six fittest, up to two protected slots for the most behaviorally unusual, and the newest arrivals. Each generation adds a batch of engineer proposals and adversary cases; the loop stops when two consecutive generations bring neither a better champion nor an accepted novel scorer. The initial population is four baseline scorers implementing standard ideas from the creativity literature: word rarity, unusual word combinations, an “is this overdone?” sanity check, and their product.

4 Experiments

Arm A (larger model, low volume). A mid-size commercial LLM played both roles for three generations: 3 scorer proposals and 8 adversarial candidates per generation (6 kept).

Arm B (smaller model, high volume). A much smaller, cheaper LLM played both roles under an identical harness, seeds, and gates, but at three times the mutation volume — 9 proposals per generation from three parallel engineers with different briefs, and 16 adversarial candidates (10 kept) from two parallel adversaries — and ran for seven agent generations. Everything else was held fixed.

Both arms start from the same 42 test cases (20 built from real internet text plus 22 from two earlier adversarial rounds), split deterministically into 29 training and 13 held-out cases.

	gen	train	champion (best scorer)	sep.	held-out	core
A	0	29	unusual-combination baseline	0.072	0.059	0.212
A	1	33	added-detail structure (agent)	0.032	0.018	−0.003
A	2	37	kept-skeleton + fresh insert (agent)	0.129	0.161	0.202
A	3	41	(same champion holds)	0.074	0.121	0.202
B	0	29	unusual-combination baseline	0.072	0.059	0.212
B	1	36	insert-vs-swap structure (agent)	0.074	0.098	0.135
B	2	43	short-word concreteness (agent)	0.201	0.190	−0.009
B	3	50	(reverts to gen-1 champion)	0.055	0.068	0.135
B	4	57	variety × rarity gate (agent)	0.106	0.158	0.359
B	5	63	(same champion holds)	0.139	0.187	0.359
B	6	70	(same champion holds)	0.123	0.169	0.359
B	7	77	(same champion holds)	0.154	0.198	0.359

Table 1: Champion trajectory in both arms. “sep.” is separation on the (growing) training set; “held-out” is separation on cases no agent ever saw; “core” is the never-growing stationary subset. Agent-written champions displaced the baselines in both arms.

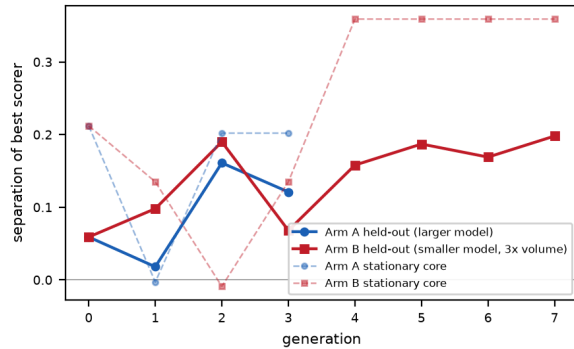


Figure 1: Held-out separation (solid) and stationary-core separation (dashed) of each generation's champion. Arm B is chaotic early — including a generation-2 champion whose core score went *negative*, i.e., it had overfit the adversarial distribution — then converges to the strongest scorer of either arm.

5 Results

5.1 A Classical Principle, Rediscovered as Code

Creativity research has long held that creativity is not novelty alone: an idea must be both *new* and *fitting* (Runco and Jaeger 2012), and preference for novelty peaks at moderate levels rather than growing without bound (Berlyne 1971). Nobody told the agents this. Yet in both arms, every surviving champion has the same architecture: a novelty signal **multiplied by a sanity gate**. Arm A's discovery scores “how novel is the material you inserted, *given* that you kept the skeleton of the original intact”; Arm B's scores “how varied is the rewrite's structure, *given* that its content earns the variety.” Single-signal scorers — rarity alone, divergence alone, concreteness alone — were each destroyed by an adversarial round within one generation of taking the lead. The two-factor multiplicative shape was not seeded and was not suggested in any prompt; it emerged twice, independently, because it is the only shape the adversary could not decouple.

5.2 Model Quality Sets the Pace, Not the Destination

	Arm A (larger)	Arm B (smaller, 3× vol.)
agent generations	3	7
scorer proposals	9	45+
accepted	3 (33%)	~15 (33%)
copycats caught by gate 3	3	12+
champion turnovers	2	4 (one regression)
final champion held-out sep.	0.121	0.198
final champion core sep.	0.202	0.359
strongest adversary round	−0.68 avg. margin worst possible (−1.0) × 10	

Table 2: Head-to-head. The held-out sets diverge across arms after generation 0 (different adversaries grew them), so the stationary core — the same six cases in both arms — is the honest cross-arm comparison.

Three observations answer Q2.

(a) The gates, not the proposer, set the quality bar.

The acceptance rate was 33% under both models. The smaller model produced four times as many “copycats” —

proposals whose stated mechanism sounded new but whose behavior correlated up to 0.94 with an existing scorer — and the behavioral-novelty gate caught every one. Claimed mechanisms do not survive contact with measured behavior.

(b) Attacking is easier than building. The smaller model's adversaries matched and then exceeded the larger model's: one of its rounds achieved the worst possible margin on all ten kept cases, the hardest single hit in either arm. Finding a scorer's blind spot appears to demand less capability than designing a scorer.

(c) Volume plus generations compensate for weaker proposals. The smaller model's path was turbulent — a champion overturned every generation for three rounds, one overfit champion exposed by the stationary core, one outright regression — but at three times the proposal volume and with four extra generations it reached a final champion that beats the larger model's on the only strictly comparable number (core separation 0.359 vs. 0.202) and survived four consecutive targeted attack rounds.

5.3 What the Honesty Machinery Caught

The run is as interesting for what the referee *rejected* as for what it kept.

A memorizing scorer. In the pilot that motivated gate 2, an agent shown the champion's failures copied their distinctive vocabulary into its own word lists; its measured skill was partly a lookup table of the test set. The automated code-inspection gate now rejects this pattern outright.

A distribution-overfit champion. Arm B's generation-2 leader posted the best training separation seen to that point (0.201) while scoring *below zero* on the stationary core: it separated the adversary-authored cases while failing ordinary prose. Without a never-growing tier, this would have read as a breakthrough.

A test-growth rule that backfired. Our rule kept the ten candidate cases the champion handled *worst*. Once the champion became strong enough that every attack failed, “worst” cases were still ranked correctly — so the test set began absorbing champion-friendly cases and the champion's measured skill inflated. An adversarial test set only stays hard while the adversary stays competitive; the keep-rule must require the champion to actually fail, or discard the batch.

5.4 The Ceiling

The loop eventually converged by *adversary exhaustion*: in Arm B's final generation, neither of two fresh attack angles produced a single case the champion mis-ranked, and the engineers' remaining textbook ideas (a standard lexical-diversity algorithm, a statistical de-confounding trick, two-word-sequence surprise) each failed the gates on the merits. The concepts that stayed unmeasured — whether an inserted image actually *fits* its context, true meaning-level distance — are exactly the ones that word-frequency statistics cannot express. One adversary

demonstrated this pointedly, submitting pairs whose two rewrites are statistically near-identical but semantically opposed (an apt image versus a subtly impossible one). No scorer in the population can tell such pairs apart, and on the evidence here, none built from frequency statistics ever will. That boundary — freshness is measurable, fit is not — is, we think, the clearest empirical statement of where interpretable text metrics end and meaning-aware models must begin.

6 From Measuring Creativity to Improving It: a Research Agenda

So far the scorers have been judges. But the original reason to want cheap, deterministic, inspectable judges is what they unlock: a function that runs in microseconds can be called millions of times inside a *training* loop — as the reward for reinforcement-learning fine-tuning (RLFT) of a text generator, or as the fitness function of an evolutionary search over generating programs. Seen from that angle, everything we learned about how scorers fail is really a lesson about how rewards get **hacked**, learned cheaply before any expensive training run. This section turns each empirical lesson into a concrete proposal.

6.1 Harden the Reward Before Training, Not During

A policy trained to maximize a naive creativity reward will discover, within hours, exactly the exploits our adversary found in minutes: flood a rarity reward with ornate vocabulary (our purple-prose attack), flood a change-based reward with meaning-preserving shuffles (synonym churn), flood an elaboration reward with empty qualifiers (padding). Each adversarial axis in our runs is a *preview of a reward-hacking failure mode*, obtained at the cost of a few agent calls rather than a training run. We therefore propose treating our loop as a standard **pre-flight procedure for reward functions**: before optimizing against any subjective-quality reward, run agent red teams against it and report its **decouplability** — how easily an attacker produces high-reward, low-quality text — with the same rigor as its agreement with human judgment. Our margin statistics give decouplability a number.

The runs also say *which reward shapes survive*. Additive rewards are decouplable by maximizing their strongest term. Multiplicative, gated rewards force an attacker to satisfy every factor simultaneously — which is precisely why both arms converged on products. Two structural rules follow for creativity rewards: **(i)** compose them as a novelty term times one or more sanity gates (appropriateness, coherence), never as a weighted sum; **(ii)** shape the novelty term as an inverted U — reward *moderate* novelty relative to the source and penalize overshoot — so that “more is always better” never holds along any single axis. The century-old observation that people prefer moderate novelty (Berlyne 1971) here becomes a practical reward-shaping principle: the peak of the U is exactly where a reward stops being gameable by exaggeration.

6.2 Reward the Edit, Not the Text

The single most portable design element our loop discovered is **relational scoring**: every strong scorer measured the rewrite *against its own source* — what was kept, what was inserted, how far the change went — rather than measuring the text in isolation. For RLFT this matters twice. First, an anchor-relative reward cannot be satisfied by collapsing to one high-scoring house style, because each prompt carries its own baseline and reward is only available by improving *this* text; style collapse is the classic failure of absolute style rewards. Second, distribution-level separation gives *dense* credit: a policy earns partial reward for partial improvement, where accuracy-style rewards are all-or-nothing.

Two refinements from late generations are directly usable as training objectives. **Novelty-per-edit** — dividing novelty gained by the size of the edit — makes one perfect inserted image outrank a wholesale rewrite, encoding an editor's economy as an optimizable quantity. And **concentration of change** — whether the novelty is localized in one or two spans or smeared across the text — distinguishes a landed image from churn using only alignment statistics. Both are cheap, deterministic, and were invented by the agents under adversarial pressure.

Finally, our ceiling result dictates a **two-tier reward**: the deterministic scorer as the dense, every-sample signal, and an expensive meaning-aware check (embedding similarity, an LLM judge, or a human) applied *sparsely*, only to candidates the cheap tier already rates highly. The boundary we located — word statistics certify freshness but cannot certify fit — is exactly the boundary where the budget should shift tiers.

6.3 Close the Loop: the Generator Is the Strongest Red Team

Our adversary writes attacks by reasoning about the champion's code. A policy under optimization does something stronger: it *searches* the reward landscape directly and finds exploits no reasoner anticipates. The natural next system therefore closes the loop (Figure 2): train or evolve a generator against the current hardened scorer ensemble; mine its highest-reward outputs; have a judge (human or meaning-aware model) label which are genuinely creative and which merely score well; and feed the high-reward-but-hollow ones back as new adversarial test cases — the best test cases obtainable, because they are the reward's *actual* failure modes. The scorer side then evolves as in this paper, and the improved reward retrains the generator. This is generative-adversarial in spirit, with two differences that matter for science: the discriminator stays *interpretable* (its updates are code diffs with named mechanisms, not weight updates), and every escalation leaves an audit trail.

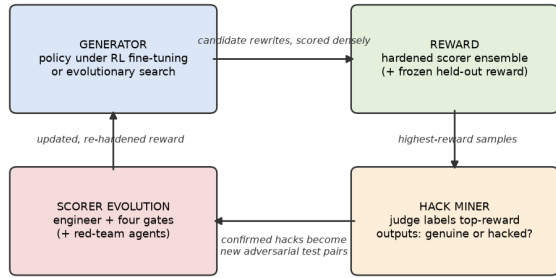


Figure 2: The proposed co-evolution loop. The right half is standard reward-driven generation; the left half is this paper’s scorer-evolution harness. The generator’s own high-reward failures become the adversarial test cases that re-harden the reward.

Our failure catalog supplies the guardrails this loop needs. Keep a **frozen held-out reward** — an ensemble never used during training — and monitor the gap between training reward and held-out reward as a live hacking meter (our stationary core, generalized). Require kept test cases to actually *defeat* the current reward, or the eval dilutes and measured skill inflates (our backfired keep-rule). And keep the reward ensemble behaviorally diverse using fingerprint decorrelation, so the policy cannot satisfy a single signal family and call it creativity.

6.4 What a Scorer Can Teach Us About Creativity Itself

The same machinery is an instrument for the science of creativity, not just its engineering. Four directions look most promising.

An empirical factor structure. The behavioral-novelty gate does something no hand-built battery does: it discovers axes of textual creativity that are *behaviorally independent by construction* — in our runs, word-level novelty, combination-level novelty, structural variety, insertion locality, and appropriateness gates. Regressing human creativity ratings onto these axes would yield a data-driven, fully interpretable decomposition of what readers mean by “creative” — which axes carry weight, which are redundant, and whether the weights differ across genres and readers. Every discovered scorer is a falsifiable hypothesis; human ratings are the experiment.

Domain transfer as a universality probe. In a side study scoring acclaimed song lyrics against generic ones, word-rarity *inverted* — plain-diction masters score low on rarity — while combination-level novelty transferred cleanly, preferring every acclaimed original over its flattened paraphrase. This suggests a testable hypothesis: *unusual combinations of ordinary words are the domain-general core of textual creativity, while rare vocabulary is a domain-specific costume.* The harness makes such cross-domain tests nearly free.

A measurement-complexity ladder. Our runs empirically locate which facets of creativity are measurable with which resources: word-frequency statistics suffice for

freshness; co-occurrence statistics for combinational surprise; embeddings are needed for semantic distance and topical fit; and judging whether a sustained image is *earned* plausibly requires a full language model. The adversary tells you when a rung is exhausted: attacks from within the level stop working, and the only effective attacks come from the level above (our statistically-identical, semantically-opposed pairs). Charting this ladder — the minimum machinery required for each judgment — would give creativity measurement the kind of resource-complexity map that computational linguistics has for syntax and semantics.

Open questions the harness can answer cheaply. Is the peak of the inverted U stable across genres and readers, or a moving target? Do humans actually prefer minimal-edit brilliance (high concentration of change), or is that an editor’s aesthetic? Does appropriateness act as a hard gate in human judgment (a conjunctive product, as our survivors assume) or as a soft trade-off? Each question reduces to one scorer-versus-human-ratings experiment on a few hundred pairs.

7 Related Work

Our loop belongs to the family of LLM-guided evolutionary search over programs, most prominently FunSearch (Romera-Paredes et al. 2024), which pairs an LLM proposer with a programmatic evaluator; we add an adversarial data-growth role and honesty gates targeted at the failure modes of subjective evaluation. Adversarially-collected benchmarks that grow with the model under test were popularized by Dynabench (Kiela et al. 2021). Program-space search by selection descends from genetic programming (Koza 1992). On the creativity-measurement side, we build on the two-factor definition (Runco and Jaeger 2012), moderate-novelty preference (Berlyne 1971), computational novelty measures based on semantic distance (Beaty and Johnson 2021; Olson et al. 2021; Johnson et al. 2022), and lexical-diversity measurement (McCarthy and Jarvis 2010). Our contribution is not a new creativity theory but evidence that a gated adversarial loop can re-derive the theory’s structure as executable code, and an ablation isolating how much the proposing model matters.

8 Limitations

The “more/less creative” labels were authored by LLMs, not humans; human ratings are the essential next validation. Each arm was run once — trajectory details (which champion falls when) are surely seed-dependent, though the convergent design shape appeared in both arms. The stationary core is only six cases. Separations are modest in absolute terms (0.1–0.4 on a 0–1 scale) because the test set is deliberately adversarial. Both arms received the same orchestrated lesson-passing between generations (each generation’s context summarizes what killed the last champion); we view this memory as part of the evolutionary machinery, but it means our ablation isolates the proposing

model, not the orchestration. Finally, pre-trained semantic resources were unavailable in our sandboxed environment, which made the frequency-statistics ceiling unavoidable rather than chosen.

9 Conclusion

A small agent ensemble — one role writing scoring code, one role writing counter-examples, and a deterministic referee enforcing validity, non-memorization, behavioral novelty, and usefulness — twice rediscovered the two-factor structure of creativity as short, readable Python, and produced scorers that survive attacks their own authors designed. Swapping the proposing model for one far smaller changed the journey but not the destination: identical acceptance rates, equal-or-better final champions, the same convergent design. The practical recipe we take away is that when searching for evaluation metrics with LLMs, **invest in the referee and the adversary before investing in a bigger proposer** — and always keep one test set the process is never allowed to touch.

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